



Education

Degree / Certificate	Institute / Board	CGPA / Percentage	Year
B.E.	Jadavpur University, Kolkata	8.97 / 10	2024 – 2028
Senior Secondary (ISC)	Don Bosco School, Liluah	92.5%	2021–2023
Secondary (ICSE)	Don Bosco School, Liluah	93.2%	2010–2021

Certificates and Courses

Course / Certification	Institute	Status
Professional Certificate Course in Data Science	IIT Kanpur	Completed [View]
Professional Certificate Program in Gen AI, ML & Intelligent Automation	IIT Delhi	Completed [View]
- Applied Data Science with Python [View] - Core Machine Learning [View] - Foundations: SQL [View]	Simplilearn	Certified

Technical Skills

Category	Tools / Technologies	Additional Info
Programming	Python • C • Java • JavaScript	Intermediate – Advanced
ML / DL	NumPy • Pandas • Scikit-Learn • PyTorch • TensorFlow	Classification, Segmentation, Anomaly Detection
Embedded Systems	Arduino • ESP32 • L298N • MPU6050 • IR Sensors	Real-time feedback & control
Tools & Platforms	Git • GitHub • Kaggle • MATLAB • Simulink	Version control, simulation

Projects

TraffiQ: Autonomous Driving System 🚗	Brain MRI Tumor Segmentation 🧠🌐Demo
- Developed an end-to-end navigation system using Behavioral Cloning and YOLOv8n for obstacle detection. - Trained on Udacity and CARLA simulator datasets; optimized via ONNX for real-time inference on Raspberry Pi 4B. <i>Stack: Python, PyTorch, ONNX, YOLOv8, OpenCV, CARLA</i>	- U-Net CNN for pixel-level MRI tumour segmentation; preprocessing, augmentation, Dice evaluation. - Deployed as an interactive diagnostic app via Streamlit for real-time visualization. <i>Stack: Python, PyTorch, OpenCV, Streamlit</i>
Multi-Agent Research Laboratory 🤖🌐Demo	Self-Balancing Robot
- Developed an advanced collaborative multi-agent platform performing real-time web search, content scraping, synthesis, and academic evaluation. - Streamed live SSE logs to a responsive, premium dashboard with direct exports to formatted PDF and DOCX. <i>Stack: Python, FastAPI, LangChain, React, TypeScript, TailwindCSS</i>	- Two-wheeled inverted-pendulum robot with tuned PID control for stable real-time balancing. - Implemented MPU6050 sensor fusion in C++. <i>Stack: C/C++, Arduino, MPU6050</i>

Achievements

Achievement	Competition / Event
Ranked in Top 5 teams (All India) for TraffiQ: Autonomous Driving System	Srijan Competition